

Curriculum Vitae

Personal Information

Name Sayeda Tashnuba Jahan
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Education

2026-Present **University of Lethbridge** *Lethbridge, Alberta, Canada*
Ph.D. in Theoretical Physics, *Quantum Horizons Alberta*
2022-2023 **University of Bologna** *Bologna, Italy*
Masters in Theoretical Physics - Achieved 2nd position in the ranking for scholarship provided by the Department of Physics and Astronomy (DIFA)
2020-2021 **Universite Paris-Saclay** *Orsay, France*
Masters (M1) in General Physics
2016 - 2019 **BRAC University** *Dhaka, Bangladesh*
Bachelor of Science in Physics GPA : 3.61/4.00

Work Experience

Jan 2020 - Oct 2020, **BRAC University** (*Dhaka, Bangladesh*) <https://www.bracu.ac.bd/> – Lecturer, Department of Mathematics and Natural Sciences (MNS).
Jan 2021 - Sept 2022,
Jan 2024 - Present

- o Launched first ever undergraduate course on black holes (PHY419); Tailored notes can be found here: **Black Holes**
- o Taught advanced physics major courses Cosmology and Astrophysics (PHY403), Theory of General Relativity (PHY413); Tailored notes of the course can be found here: **Cosmology and Astrophysics, Theory of General Relativity**
- o Taught courses on Differential Calculus (MAT110), Integral Calculus and Differential Equations (MAT120), Electromagnetism and Special Theory of Relativity (PHY112), Thermal Physics and Radiation (PHY114), Waves, Oscillations and Acoustics (PHY113).

Feb 2021 - Jul 2021 **Institut d'Astrophysique Spatiale (IAS)** (*Orsay, France*) – Research Intern - "Estimating dark matter halo masses of Planck high starforming galaxy protoclusters, and predictions for Euclid"

Academic Achievements (Undergraduate at BRAC University)

- Vice Chancellor's Certificate (44th Batch of the Residential Semester(RS) of BRAC University).
- Dean's list in 3 Semesters.

Extra-curricular Works

- o **Reviewer at Modern Physics Letters A (MPLA)**
- o **Organizer of the Jamal Nazrul Islam Memorial Winter School (flagship event of Physics Without Frontiers programme by International Centre for Theoretical Physics (ICTP)) held in January 2025 (Website)**
- o **Organizer of Merul Badda Talks in Math and Physics (Website, Facebook)**

- o **Organizer of SPARQ** (Facebook)
- o **Other Extra-curricular Works**
 - Primary Organizer and Teaching Assistant of the School on Quantum Field Theory and the Standard Model under ICTP Physics Without Frontiers: Physics for Bangladesh. ([Link to event](#))
 - Primary Organizer and Teaching Assistant of the School on General Relativity and Cosmology under ICTP Physics Without Frontiers: Physics for Bangladesh. ([Link to event](#))
 - Conducted pre-school lectures on General Relativity and Cosmology for the Jamal Nazrul Islam Memorial Winter School 2025. ([Link of the lecture notes](#))
 - Wrote a review article on String Gas Cosmology for course titled "Quantum Cosmology". ([Link of the article](#))
 - Attended the workshop titled "New Pathways in the Exploration of Quantum Field Theory and Quantum Gravity" hosted by the International Centre for Theoretical Physics (ICTP) ([Link to event](#))
 - Hosted an online workshop under IEEE AEES titled "Observational Cosmology" for undergraduate students in BRAC University. ([Link to event](#))
 - Attended the 2022 Spring School on Superstring Theory and Related Topics held physically at the International Centre for Theoretical Physics (ICTP). ([Link to event](#))
 - Attended the 17th Modave Summer School in Mathematical Physics held in Brussels, Belgium. ([Link to event](#))

Research Interests

Ongoing Project(s): Co-PI, BRAC University (Research Seed Grant Initiative): Research on time-dependent string-theory backgrounds, developing GKP-inspired solutions in Type IIB string theory with applications to inflation, reheating, and possible primordial black hole formation.

Research Paper(s): 1. Effect of moduli redefinitions on fibre inflation ([Link of the article](#))

Undergraduate Thesis Topic: Quantization of Blackhole Horizon Area and Multipartite Entanglement of Blackhole Information Subsystems. **Supervisor:** Professor Dr. Mahbub Alam Majumdar. ([Link of the article](#))

Masters Thesis Topic: On the Quantum Nature of Primordial Black Holes from Inflationary Cosmology **Supervisor:** Professor Dr. Roberto Casadio ([Link of the article](#))

Internship Report: Estimating photometric redshifts and stellar masses of a few Planck high star-forming galaxy protoclusters with CFHT data, and predictions for Euclid **Supervisor:** Professor Hervé Dole ([Link of the article](#))

Current Research Interests: String Cosmology, Primordial Blackholes, Quantum Nature of Blackholes, Cosmological Perturbation Theory, Early Universe Cosmology, Black Hole Information Paradox.

Languages

Bangla, *Native*

English, *Advanced*

French, *Beginner*

Italian, *Beginner*

References

Professor Dr. Mahbub Alam Majumdar Dean, School of Data and Sciences, BRAC University **email:** majumdar@bracu.ac.bd

Professor Dr. Roberto Casadio Professor, Department of Physics and Astronomy, Alma Mater Studiorum - Università di Bologna. **email:** roberto.casadio3@unibo.it

Professor Dr. Michele Cicoli Professor, Department of Physics and Astronomy, Alma Mater Studiorum - Università di Bologna. **email:** michele.cicoli@unibo.it